Pragya Jha

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EDUCATION

SRM University | *Sonipat, Haryana*
Computer Science and Engineering (with specialization in Data Science and Artificial Intelligence) | ***B.Tech***

TECHNICAL SKILLS

- **Languages:** Python, R
- **Backend & Frameworks:** Flask, FastAPI, REST APIs
- **Data Science & ML:** Data Science, Machine Learning, Artificial Intelligence (AI), Generative AI, Pattern Recognition, Data Visualization
- **Databases:** SQL, MySQL, MongoDB, NoSQL
- **DevOps & Tools:** Git, CI/CD, Postman, IBM Cloud
- **Core Concepts:** Data Structures & Algorithms, Software Engineering, API Testing, Cloud Application Development, Agile,Scrum

EXPERIENCE

Infosys Springboard | AI & Data Science Intern ***Oct 2025– Dec 2025***

- Developed a full-stack student analytics platform using FastAPI and Next.js, enabling tracking of 1,000+ study sessions and delivering personalised performance insights.
- Created 12+ REST APIs for managing study sessions, student data, OTP-based authentication, and admin analytics workflows.
- Built a machine learning pipeline using K-Means clustering (K=3) on 5+ behavioural features to segment students into performance categories, improving identification of at-risk students.
- Generated a rule-based recommendation engine with 10+ decision rules, generating actionable insights to improve study efficiency and reduce distractions.
- Enabled data processing and visualisation pipelines using Pandas, Scikit-learn, and charting tools, supporting real-time dashboards and reducing analysis time by ~30%.

Edunet Foundation (AICTE & IBM SkillsBuild) | *AI & Cloud Intern* ***July 2025– Aug 2025***

- Assembled an LLM-powered conversational AI agent using IBM watsonx.ai, implementing LangGraph-based orchestration with ReAct architecture for reasoning and tool invocation.
- Integrated Granite-3-3-8B-Instruct LLM with optimised decoding parameters, enabling context-aware itinerary generation across 500+ multi-turn interactions.
- Executed tool-augmented pipelines for travel planning (itinerary generation, destination recommendations, weather-aware adjustments) by integrating 3+ external APIs.
- Architected a stateful multi-turn conversation system with orchestrated decision-making, improving response relevance and task completion across complex user queries.
- Engineered modular Python-based tool calling and workflow orchestration, enabling scalable and extensible travel planning with structured outputs and <400 ms response latency.

Xyronix Labs | *SDE Intern* ***Jan 2025– July 2025***

- Built and optimised RESTful APIs using Flask, improving response time by 30%.
- Collaborated on scalable cloud-based solutions integrating OAuth2 services.
- Partnered with frontend and DevOps teams to integrate APIs into live systems.

KEY PROJECTS

Emotion-Aware Wellness Chatbot | Tech stack: **Python, Hugging Face Transformers, Groq LLM** **February 2026**

- Established an NLP-driven chatbot using transformer-based models (BERT/RoBERTa) to classify user sentiment and emotions across 5+ categories with real-time inference.
- Applied Groq LLM for context-aware response generation, reducing irrelevant responses and improving personalisation quality across 1,000+ interactions.
- Designed a modular inference pipeline (emotion detection → context mapping → response generation), reducing end-to-end response latency to <300 ms.
- Implemented session-based context handling and prompt engineering techniques to maintain conversational continuity and improve response coherence.
- Structured an end-to-end system delivering personalised coping strategies and mood tracking, increasing user engagement and retention in simulated test scenarios.

Exoplanet Data Analysis & Visualisation | *Tech Stack:* ***Python, Pandas, Matplotlib, Seaborn*** ***October 2025***

- Performed data preprocessing and feature engineering on NASA exoplanet datasets (10,000+ records), handling missing values, normalisation, and outlier detection.
- Conducted exploratory data analysis using statistical methods and correlation mapping to identify relationships between planetary features (mass, radius, orbital parameters).
- Programmed advanced visualisation pipelines (multi-dimensional plots, distribution analysis, and habitability zone mapping) to extract and present key insights.
- Crafted a modular data pipeline (data ingestion → preprocessing → analysis → reporting), improving reproducibility and scalability of experiments.
- Applied time-series and feature correlation analysis to uncover trends and validate hypotheses in exoplanet discovery patterns.

CERTIFICATIONS

- **Cloud Application Development –** *IBM* (IBM Cloud, REST APIs, Microservices Architecture, Cloud Deployment)
- **Machine Learning using R –** *IBM* (Supervised Learning, Regression & Classification, Statistical Modeling, Model Evaluation)